

Solution: The set of positive integers x can either be: 1) $x < 5$ or 2) $x \geq 5$

Case 1: $x < 5$. $1^3 = 1 \neq 100$ $2^3 = 8 \neq 100$ $3^3 = 27 \neq 100$ $4^3 = 64 \neq 100$

Case 2: $x \geq 5$. $5^3 = 125 > 100$. It's easy to see that for all $x \geq 5$, $x^3 > 100$.

$\therefore 100$ is not the cube of a positive integer $\square$

4. Prove that there are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers.

Solution:

~~Perfect cubes less than 1000~~

$1 = 1^3$. $8 = 2^3$. $27 = 3^3$. $64 = 4^3$. $125 = 5^3$. $216 = 6^3$. $343 = 7^3$
 $512 = 8^3$. $729 = 9^3$

We shall now take each pair of cubes and check whether their sum gives a perfect cube less than 1000.

- $1+8=9$, $1+27=28$, $1+64=65$, $1+125=126$, $1+216=217$, $1+343=344$,
 $1+512=513$, $1+729=730$, $1+1=2$
- $8+8=16$, $8+27=35$, $8+64=72$, $8+125=133$, $8+216=224$, $8+343=351$,
 $8+512=520$, $8+729=737$
- $27+27=54$, $27+64=91$, $27+125=152$, $27+216=243$, $27+343=370$,
 $27+512=539$, $27+729=756$
- $64+64=128$, $64+125=189$, $64+216=280$, $64+343=407$, $64+512=576$,
 $64+729=793$
- $125+125=250$, $125+216=341$, $125+343=468$, $125+512=637$,
 $125+729=854$
- $216+216=432$, $216+343=559$, $216+512=728$, $216+729=945$
- $343+343=686$, $343+512=855$, $343+729=1072$
- $512+512=1024$, $512+729=1241$ $729+729=1458$